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Classifying *E. coli* and *S. aureus* with CNNs Across Different Lab Settings

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Abstract

Motivation: This study evaluates the practicality of training convolutional neural networks (CNNs) to classify highly homogenized microscopy images generated within a single lab. It further investigates how dataset size and differences in lab environments impact model performance, comparing training a CNN from scratch with fine-tuning a pretrained model.
Results: While all models achieved near-perfect accuracy on internal validation data, their performance varied dramatically when evaluated on external datasets originating from a different lab environment. As expected, the larger Roboflow dataset (~500 images) enabled better generalization than the smaller, augmented DIBaS dataset (~40 images) when used to fine-tune a pretrained ResNet18 model. This pattern held true for both the ResNet18 models and simple CNNs trained from scratch. The Roboflow-trained ResNet18 achieved the best external performance, reaching approximately 83% accuracy on both *E. coli* and *S. aureus* when tested on the DIBaS dataset. In contrast, the DIBaS-trained ResNet18 showed highly imbalanced predictions (90% for *E. coli*, 20% for *S. aureus*) on the Roboflow test set. The simple CNNs performed worse overall with the Roboflow-trained CNN displaying extreme class bias, predicting *E. coli* over 99% of the time, while the DIBaS-trained CNN showed improved class balance (81% for *E. coli*, 32% for *S. aureus*). Although simpler models trained more quickly, they generalized poorly. These results highlight that both dataset size and model architecture play critical roles in achieving robust performance across different lab environments, supporting the use of larger datasets and pretrained networks for reliable bacterial image classification.

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1 Introduction

Traditionally, manually classifying hundreds of microscopy images of bacteria or other model organisms has been a labor intensive task which is prone to human error. In general research settings, this can serve as a bottleneck that slows down experimental workflows, even reducing the reproducibility of results. Recent advancements in computer vision such as convolutional neural networks (CNNs) offer a very promising alternative. CNNs are able to extract features from images and be trained to recognize visual patterns with high accuracy. It has started becoming more common to use similar models to automate image classification tasks in many lab workflows. Using previously manually classified images as training data, these models can dramatically reduce and potentially remove the bottleneck of manual image analysis.

This project focuses on applying CNNs to classify images of two commonly studied bacterial species in microbiology: *Escherichia coli* (E. coli) and *Staphylococcus aureus* (S. aureus), which differ morphologically as rod and circle shaped cells, respectively. The central aim is to evaluate how practical and accurate CNN-based classification can be when working with image datasets generated in real lab conditions. Specifically, this study models two realistic scenarios that a research lab might face when collecting data for model training. One dataset, sourced from Roboflow, contains 243 images of *E. coli* and 309 images of *S. aureus* and represents a situation where a lab has the ability to generate a relatively large and homogeneous set of images. The second dataset, from DIBaS, originally included only 20 images per class and was expanded to 520 images per class using image augmentation techniques. This is meant to model a more constrained setting where only a small amount of data is available and must be artificially scaled.

In addition to comparing these two datasets, this project also evaluates the effect of model choice. A key question is whether using a pretrained model (specifically ResNet18) offers advantages over a simpler CNN model trained from scratch. The pretrained model used in this study, ResNet18, has already learned many common image features from large image datasets and could potentially have improved generalization even with limited training data. Meanwhile simpler models can be a less computationally expensive alternative if there is enough data available.

The hypothesis of this study is that the Roboflow dataset, due to its greater number and diversity of images, combined with the pretrained ResNet18 model, will yield the highest performance, especially in terms of generalization accuracy. This would suggest that pretrained models are a more practical and effective solution when sufficient training data is available, while also evaluating whether smaller, augmented datasets with simpler models can serve as a reasonable fallback in more resource-limited settings.

2 Methods

1. Dataset Preparation and Preprocessing

This study used two image datasets: one from Roboflow and the other from the DIBaS collection [1]. The Roboflow dataset included 243 images of *Escherichia coli* and 309 images of *Staphylococcus aureus*, all generated under consistent imaging conditions, making it highly homogeneous. The DIBaS dataset, as described by Shaily and Kala [1], originally contained only 20 images per class. Following their methodology, this dataset was augmented to 520 images per class using a combination of horizontal and vertical flipping as well as affine transformations.

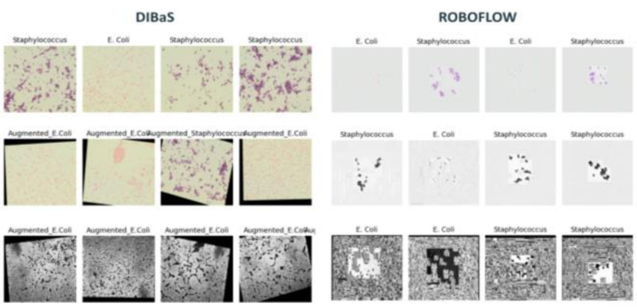


Fig. 1 DIBaS and Roboflow image examples and transforms. Each dataset shows the original images on top. DIBaS shows the augmentation by flipping followed by grayscale and histogram equalization while Roboflow shows grayscale followed by histogram equalization. The bottom images are the ones used for training each model.

All images underwent the same preprocessing pipeline. First, they were converted to grayscale to eliminate irrelevant variation due to staining or dye differences. Then, histogram equalization was applied using OpenCV to standardize brightness and contrast. Each image was resized to 224×224 pixels to match the expected input size of ResNet18. Pixel values were normalized to the range [−1, 1]. For the Roboflow dataset, additional augmentations such as random flipping and translation were applied during training to reduce overfitting to certain features such as bacterial location.

2. Model Architectures and Training

Four convolutional neural networks (CNNs) were trained in this study to classify microscopy images of bacteria. Two of these models were based on a pretrained ResNet18 architecture from PyTorch, while the other two were custom CNN models trained from scratch following the pytorch image classification tutorial for CNNs.

The first model was a ResNet18 pretrained on ImageNet and fine-tuned on the Roboflow dataset. The first convolutional layer was modified to accept a single-channel image to take the grayscale images as input. The final fully connected layer was also replaced with a new layer outputting two logits, corresponding to the two bacterial classes. The second model used the same modified ResNet18 architecture but was instead fine-tuned on the augmented DIBaS dataset. For both models, the Adam optimizer was used with a learning rate determined through manual sweep testing, and the loss function was set to PyTorch's CrossEntropyLoss(). Each model was trained for 5 epochs because after 5 epochs training loss begins to plateau and fluctuate. Both models were evaluated not only on their internal validation folds but also on the external dataset (i.e., DIBaS for the Roboflow-trained model, and vice versa).

The other two models were simple CNNs designed and trained from scratch (one trained on DIBaS and the other trained on Roboflow). These models included two convolutional layers (with kernel size 5), each followed by a ReLU activation and max-pooling, and three fully connected layers. The first convolutional layer took in a 1-channel image and output 6 feature maps. The second convolutional layer accepted those 6 channels and output 16. After flattening, the first fully connected layer mapped to 120 features, the second to 84, and the final to 2 output logits for classification. The model was trained using the same optimizer and loss function as the ResNet18 models. The Roboflow CNN model was trained on 15 epochs as the training loss did not plateau at 5.

3. Evaluation and Cross-Validation

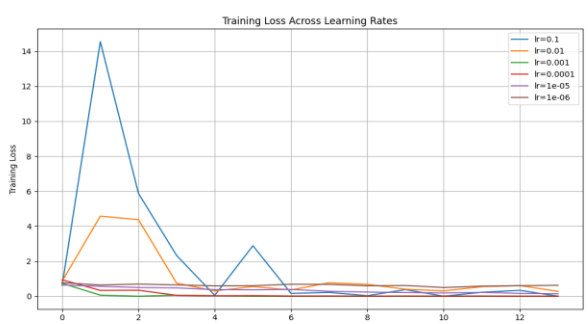
All models were trained and evaluated using 5-fold cross-validation. In each fold, 80% of the dataset was used for training and 20% for validation. The best-performing model from each fold, as determined by validation accuracy, was saved using PyTorch. A batch size of 32 was used for all training runs. For learning rate optimization, a sweep was conducted across several orders of magnitude (from 1e-1 to 1e-5) observing which learning rate led the model to converge the fastest based on training loss over batches.

To evaluate generalization performance, each model was tested on the external dataset not used in its training. Specifically, the models trained on Roboflow were tested on the augmented DIBaS dataset (520 images per class), while the DIBaS-trained models (both the ResNet18 and the simple CNN) were tested on the full Roboflow dataset. For final performance metrics, both accuracy and F1-score were computed using scikit-learn's f1_score function to account for class balance.

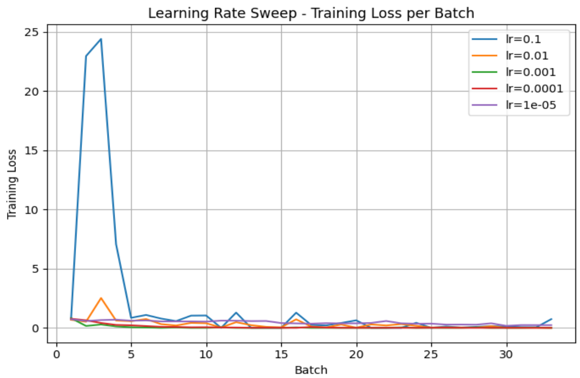
3 Results

1. Learning Rate Sweep and Training Curves

ResNet18 with Roboflow



ResNet18 with DIBaS



Simple CNN with Roboflow

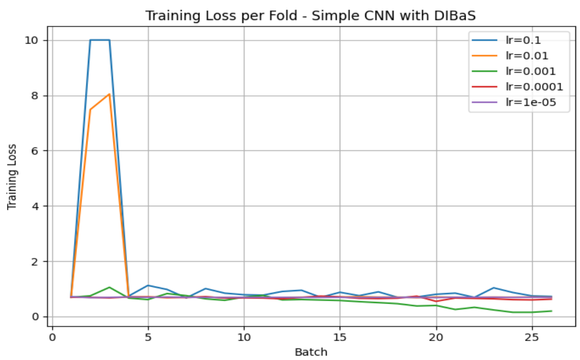
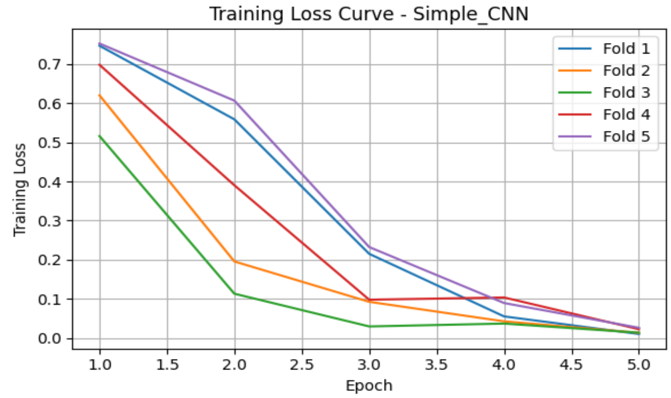
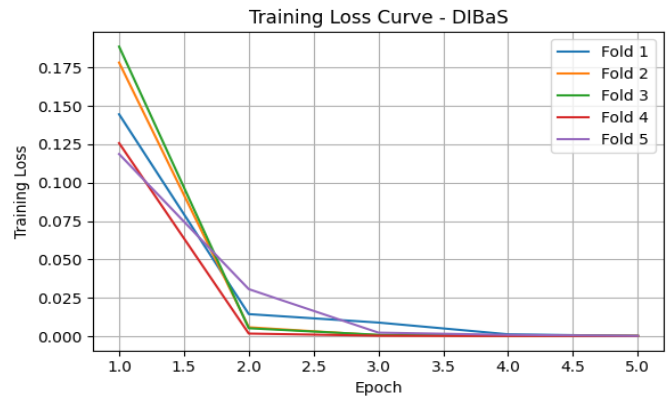


Fig. 2 Learning rate sweep across five values: 1e-1, 1e-2, 1e-3, 1e-4, and 1e-5. The optimal learning rate was selected based on which value led to the steepest drop in training loss.

For each model and dataset, a learning rate sweep was performed and used to guide the training process. All models were trained for 5 epochs except for the simple CNN trained on the Roboflow dataset which had a significantly higher loss and thus was trained on 15 epochs. Training loss curves were also generated for each fold during cross-validation, providing insight into overfitting or underfitting across folds as well as determining the number of epochs each model would be trained with. All folds were plotted for each model except for the simple CNN trained on DIBaS, which was added later.





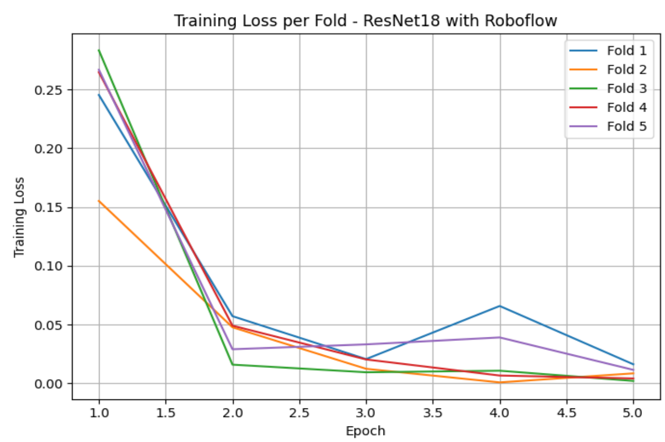
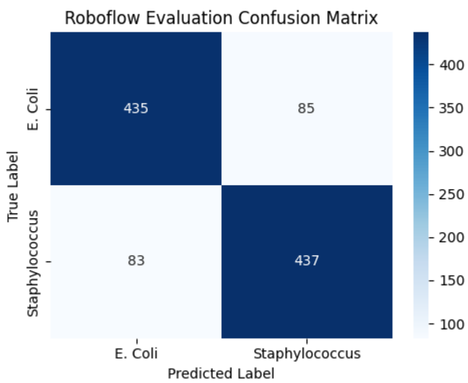


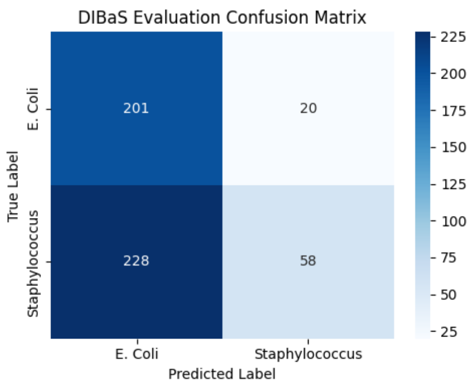
Figure 3. Training loss curves for each model across 5 epochs with a 0.001 learning rate. This figure was used to determine that 5 epochs was sufficient for all these models. The simple CNN with DIBaS was omitted as it was trained later on however it showed significantly higher loss leading to it being trained with 15 epochs instead.

2.Model Evaluation

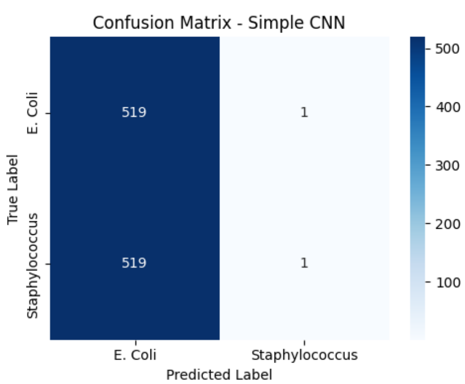
ResNet18 with Roboflow



ResNet18 with DIBaS



Simple CNN with Roboflow



Simple CNN with DIBaS

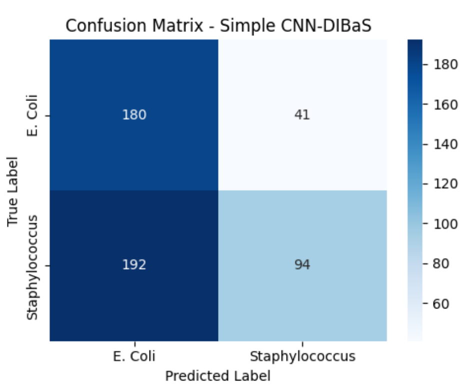


Fig. 4 Confusion matrices of each of the four models showing the ratio of correctly and incorrectly predicted labels for the external dataset. Models trained on the Roboflow dataset were evaluated based the augmented DIBaS dataset containing 520 images per class. Models trained on the DIBaS dataset were evaluated with the Roboflow dataset containing ~500 images in total.

Confusion matrices for the internal test set were omitted due to all models performing >98% accuracy. ResNet18 with Roboflow dataset performed by far the best on the external test set.

1. ResNet18 Trained on Roboflow

The pretrained ResNet18 model fine-tuned on the Roboflow dataset achieved nearly perfect performance during 5-fold cross-validation, with both training and validation accuracy reaching 100% by epoch 5 (similarly to the other models). When evaluated on the external DIBaS dataset, it showed the strongest generalization of all models, achieving 83.65% accuracy (F1: 0.9110) for *E. coli* and 84.04% accuracy (F1: 0.9133) for *S. aureus*. These results demonstrate that fine-tuning a pretrained model on a large, homogeneous dataset can yield robust performance even when tested on data collected under different lab conditions.

2. ResNet18 Trained on DIBaS

The same ResNet18 architecture, when fine-tuned on the smaller but augmented DIBaS dataset, performed well on internal validation (100%) but generalized poorly. When tested on the Roboflow dataset, the model achieved 90.95% accuracy (F1: 0.9526) for *E. coli*, but only 20.28% accuracy (F1: 0.3372) for *S. aureus*. This significant class imbalance suggests that the model overfit to the limited training distribution. Interestingly, different cross-validation folds of this model exhibited opposite biases with some favoring *S. aureus* instead of *E. coli*, indicating instability in generalization.

3. Simple CNN Trained on Roboflow

The simple CNN trained from scratch on the Roboflow dataset achieved high internal performance (~98%), but displayed extreme class bias on the external DIBaS test set. It correctly classified *E. coli* with 99.81% accuracy (F1: 0.9990) but failed almost entirely on *S. aureus*, achieving just 0.38% accuracy (F1: 0.0077). Despite being trained on a relatively large dataset, this model lacked the depth of ResNet18 and defaulted to predicting a single class, revealing the limitations of simple architectures for generalization.

4. Simple CNN Trained on DIBaS

The simple CNN trained on the augmented DIBaS dataset achieved more balanced generalization on the Roboflow test set. It reached 81.45% accuracy (F1: 0.8978) for *E. coli* and 32.87% accuracy (F1: 0.4947) for *S. aureus*. Although this performance lagged behind that of the pretrained models, it was notably better than the Roboflow-trained CNN in terms of class balance. The improved performance may be due to the stronger visual contrast and diversity introduced through augmentation, though why it generalized better remains unclear.

Conclusion

All models performed exceptionally well on their own internal validation sets, achieving near-perfect accuracy and F1 scores regardless of architecture or dataset. However, these internal results did not reliably predict performance on external datasets collected under different lab conditions. Substantial drops in external accuracy and class balance, especially in F1 scores, highlight that internal validation alone is not a sufficient measure of practical utility.

Overall, the results support the conclusion that fine-tuning a pretrained model on a large, diverse dataset (like Roboflow) yields the most robust generalization. The Roboflow-trained ResNet18 consistently outperformed other models on external tests, demonstrating high accuracy and balanced predictions across both *E. coli* and *S. aureus*. Surprisingly, the DIBaS-trained simple CNN generalized better than expected. It even outperformed its Roboflow-trained counterpart in external class balance despite being trained on a smaller, augmented dataset and a simpler architecture. This suggests that under certain conditions, data augmentation and minimal model complexity can still offer meaningful external performance.

These findings underscore an important principle for real-world lab applications: when images are uniform and originate from a consistent lab environment, even simple models trained from scratch can achieve excellent performance. However, this reliability breaks down when models are tested on data that diverge even slightly from their training distribution, such as images produced using different lab protocols, microscopes, etc. Under these conditions, the choice of both dataset composition and model architecture becomes critical. While simple CNNs offer fast and effective solutions within a fixed setting, pretrained architectures like ResNet18, especially when fine-tuned on larger, diverse datasets, demonstrated the most robust generalization across variable lab environments

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Conflict of Interest: none declared.

References

T. Shaily and S. Kala, "Bacterial Image Classification Using Convolutional Neural Networks," *2020 IEEE 17th India Council International Conference (INDICON)*, New Delhi, India, 2020, pp. 1-6, doi: 10.1109/INDICON49873.2020.9342356